Joseph Matthew Rich



jmrich@caltech.edu | joseph-rich.com | github.com/josephrich98 | Google Scholar | Los Angeles, CA

MD/PHD • COMPUTATIONAL BIOLOGY • MEDICAL IMAGING • ARTIFICIAL INTELLIGENCE • APPLIED MATH

EDUCATION

PhD (Computational Biology), *California Institute of Technology (Caltech)* May 2023 – Apr. 2027
- **Advisor**: Lior Pachter | **Dissertation**: Computational Methods and Machine Learning for Radiogenomics | **Publications**: 23 (17 first-author)

MD, *Keck School of Medicine of the University of Southern California (USC)* Aug. 2021 – Jun. 2029
- **Grants**: Beckman Institute Neuro Mini-Grant, Merkin Institute Spark Grant, AI West Med Conference Grant, RSNA Medical Student Grant

BS (Molecular, Cell and Developmental Biology), *University of California, Los Angeles (UCLA)* Sept. 2016 – Jun. 2020
- **Minor**: Biomedical Research | **GPA**: 3.99/4.0 | **MCAT**: 524/528 ("100th percentile") | Founder, Medical Literature Society | Alumni Scholar

RESEARCH EXPERIENCE

California Institute of Technology | Multimodal AI & Bioinformatics | Advisor: Lior Pachter, PhD Sept. 2023 – Present
- Developing methods to interpret AI latent spaces, with applications to neurodegenerative disease
- Developing computational and machine-learning methods that connect radiologic imaging with genomic data — RSNA 2026 oral presentation
- Troubleshooting high cross-strain contamination in mouse RNA-sequencing data for the study of RNA communication in the brain
- Benchmarked single-cell foundation models under different preprocessing for cell-type prediction — NeurIPS ICBINO-BIO workshop
- Created *biocartoon*, a free, open-source scientific illustration database and figure editor
- Created *conference-agent*, a database for keeping up to date with academic conferences
- Developed *paperpush*, a tool to automatically check academic manuscripts for journal formatting and fill out submission portals
- Created *ctkit*, a standardized CT preprocessing pipeline that removes multi-institutional confounders (TCGA/CPTAC) — JIIM (pending)
- Conducted analysis of single-cell RNA-sequencing data for a prospective clinical trial of CAR T-cell therapy in breast cancer
- Co-developed *cellsweep*, a tool that uses expectation-maximization to remove noise in scRNA-seq counts — BMC Bioinformatics (pending)
- Built *fastQpick*, a tool for sampling FASTQ reads with or without replacement for scalable subsampling and bootstrapping — Bioinformatics
- Developed *wompwomp*, a graph theory algorithm for reducing edge crossings in alluvial plots with phylogenetic heuristics — TVCG (pending)
- Developed *varseek*, a pipeline for detecting and analyzing cancer-associated variants in RNA-seq data — Nucleic Acids Research (pending)
- Compared the algorithms implemented in Seurat and Scanpy, two scRNA-seq workflow libraries, amassing 500k+ views on X — Cell Systems

USC Keck School of Medicine | Radiology AI | Advisors: Vinay Duddalwar, MD; Assad Oberai, PhD Jun. 2022 – Present
- Developing RADAR, a multimodal, agentic AI assistant for renal mass management — USC Radiology Grant Rounds oral presentation
- Conducted cost-effectiveness analysis of kidney cancer detection radiology AI models using Markov microsimulation — Urology Practice
- Evaluated deep-learning-based renal tumor image imputation models on clinically relevant questions — PLOS Digital Health
- Developed deep learning models for CT bone lesion segmentation with convolutional neural networks — European Journal of Radiology: AI

Children's Hospital Los Angeles | Radiology | Advisor: Amit Sura, MD, MBA Feb. 2023 – Present
- Performing multi-institutional retrospective and prospective analysis of AI detection for scoliosis with The Hospital for Sick Children
- Conducting a systematic review of AI in pediatric radiology — CHLA Radiology Grand Rounds oral presentation
- Identified methods to improve sensitivity of chronic lung disease of prematurity early diagnosis — SPR 2024 oral presentation, MHNP

TEACHING/WRITING EXPERIENCE

Personal Blog, *joseph-rich.com/blog*: AI, genomics, medical imaging, software development Jun. 2026 – Present

MCAT Master Tutor, *MedSchoolCoach* Apr. 2020 – Nov. 2025
- Delivered 750+ hours of individualized MCAT instruction to 30+ students. Topics included physics, general chemistry, organic chemistry, biochemistry, physiology, psychology, sociology, and critical reading. Promoted to Master Tutor for strong client ratings.

CERTIFICATIONS, SKILLS & INTERESTS

- **Mathematics & statistics**: probability, statistical modeling, biostatistics, Bayesian statistics, numerical methods, mathematical modeling, optimization, real analysis, information theory, graph theory, dimensionality reduction, microsimulation, reproducibility
- **AI & data science**: artificial intelligence, deep learning, computer vision, convolutional neural networks, vision transformers, foundation models, large language models, multimodal AI, agentic AI, PyTorch, scikit-learn
- **Computing**: Python, R, SQL, Bash, Git, Docker, AWS, REST/GraphQL APIs, web scraping, data visualization, software engineering

- **Molecular biology**: single-cell RNA-sequencing (scRNA-seq), bulk RNA-sequencing, whole genome/exome sequencing (WGS/WXS/WES), spatial transcriptomics (ST), microarray, fluorescence microscopy, western blot, polymerase chain reaction (PCR)